Text appears word by word, just like a human typing.



How It Works: Two Sides

Server side

- Writes each piece of data to the connection as soon as it is ready
- Keeps the connection **open** until all data is sent

Client side

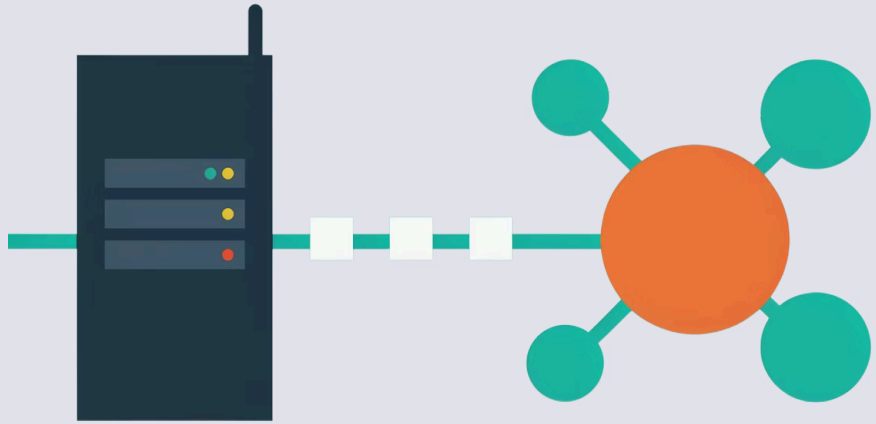
- Reads each incoming chunk as it arrives
- Displays it immediately

The Technology Behind It

Server-Sent Events (SSE): lets a server push data to a client over a long-lived HTTP connection

ReadableStream API: lets the browser read and display each chunk as it arrives

Server pushes, client reads progressively.



How It Feels to the User

Approach	Time to First Word	Total Time	User Experience
Without streaming	8 seconds	8 seconds	Long blank wait, then a wall of text
With streaming	~0.2 seconds	8 seconds	Immediate feedback, text flows naturally



Important: Total Time Doesn't Change

Streaming does not make the server faster

It lowers perceived latency by showing progress immediately

> **Same 8 seconds, very different experience.**

Key Concept: Server-Sent Events (SSE)



A simple HTTP protocol where the server pushes data to the client



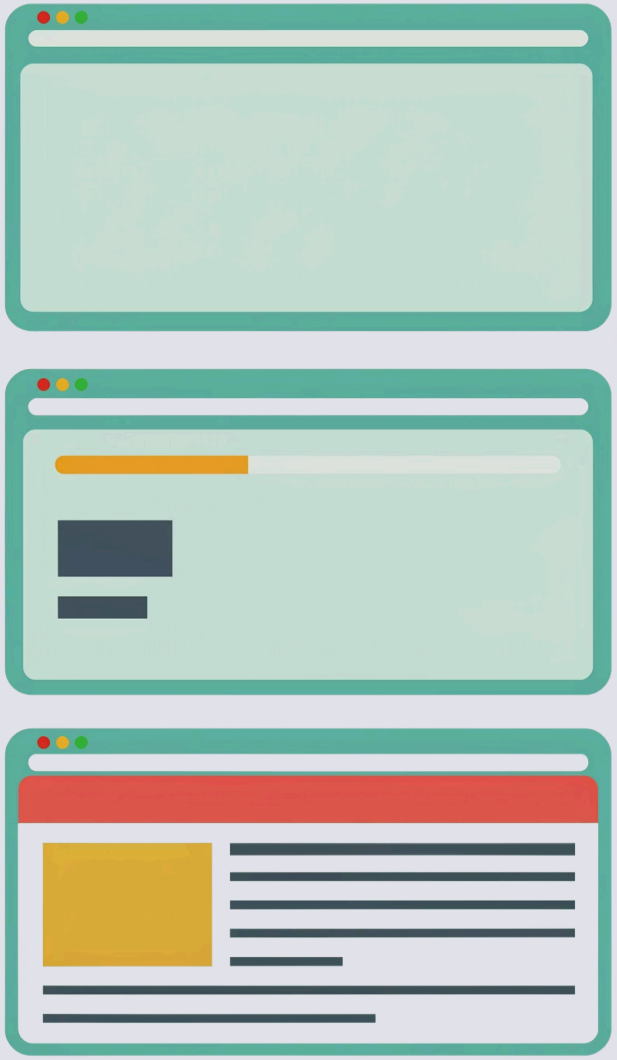
Uses a long-lived connection



Each piece is sent as a small message the moment it is ready



Best for one-way communication (server → client)



Key Concept: ReadableStream

A browser (Web) API

Lets the client read a response body piece by piece

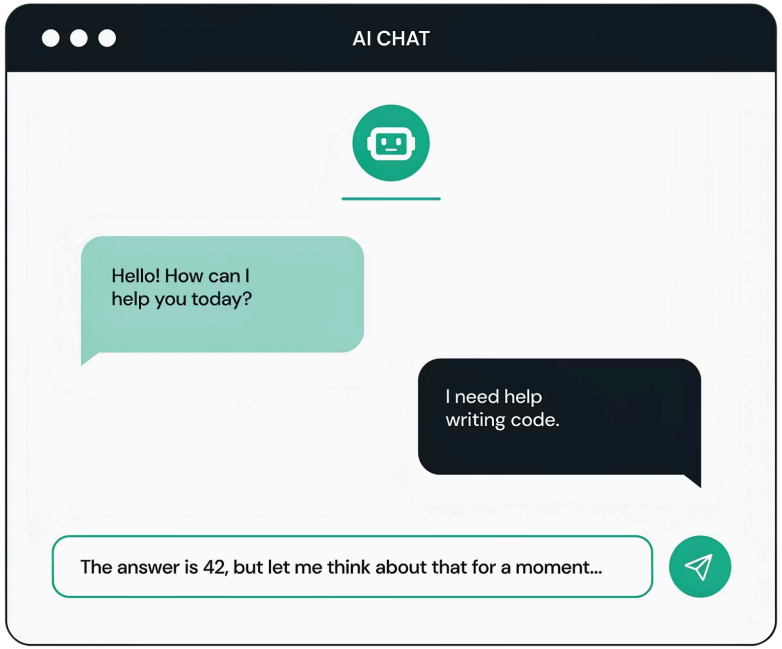
No need to wait for the entire response

This is what makes progressive rendering possible on the frontend

Key Concept: Tokens vs Words

Key Concept: Tokens vs Words

- Real AI systems generate **tokens** (roughly parts of words), not full words
- Each token is streamed as it is produced
- That's why you sometimes see a word appear **mid-letter** in AI chat interfaces



A Minimal Code Hint

The server sets one special header and writes data in small pieces: